

UI Style Replication Guide

How to faithfully recreate the Lemma AI interface in a new project

This guide documents the complete visual and interaction language of the Lemma AI front end so it can be reproduced in another codebase. The aesthetic is a **monochrome neutral (zinc) system**: light-gray canvases with floating white panels, near-black primary actions, generous 14–18 px corner radii, the Inter typeface, and lucide line icons drawn at a refined 1.75 stroke weight. The stack is **React 19 + Vite + TypeScript + Tailwind CSS v4 + shadcn/ui (new-york, neutral)**.

At a glance

Aspect	Summary
Design language	Monochrome neutral (zinc). Floating white panels on a zinc-100 canvas.
Primary action color	Near-black (zinc-900), not a saturated brand hue.
Framework	React 19, Vite 8, TypeScript 5.9.
Styling	Tailwind CSS v4 (CSS-first @theme, no JS config).
Components	shadcn/ui — style: new-york, baseColor: neutral, CSS variables on.
Primitives	Radix UI (via the radix-ui meta package + Slot).
Class helper	cn() = clsx + tailwind-merge; variants via class-variance-authority (cva).
Typeface	Inter (falls back to system UI sans-serif).
Icons	lucide-react; size 18px in nav, strokeWidth 1.75.
Data / state	TanStack Query v5 for server state; Zustand for UI state.
i18n	i18next + react-i18next; default en, zh secondary, flat keyed namespaces.
Motion	tw-animate-css + small custom CSS keyframes; hidden/fade scrollbars.

1. Project scaffolding

Create a Vite React-TS app and install the dependency set that produces this look. Versions are the ones in use; pin to them to guarantee identical behavior.

Package	Version / note
react / react-dom	^19.2
vite	^8.0
typescript	~5.9
tailwindcss	^4.2 (+ @tailwindcss/vite)
tw-animate-css	^1.4
class-variance-authority	^0.7
clsx + tailwind-merge	^2.1 / ^3.5
radix-ui	^1.5 (meta package)
lucide-react	^0.577
@tanstack/react-query	^5.91
zustand	^5.0
i18next + react-i18next	^26 / ^17
sonner	^2.0 (toasts)
inter font	@fontsource/inter or Google Fonts

Path aliases

The whole code base uses the `@/*` alias pointing at `src`. Configure it in both `tsconfig` and `vite.config.ts` so imports resolve identically.

`vite.config.ts`

```
// vite.config.ts
import { defineConfig } from 'vite'
import react from '@vitejs/plugin-react'
import tailwindcss from '@tailwindcss/vite'
import path from 'node:path'

export default defineConfig({
  plugins: [react(), tailwindcss()],
  resolve: {
    alias: { '@': path.resolve(__dirname, './src') },
  },
})
```

2. Design tokens (the heart of the look)

Tailwind v4 keeps all tokens in CSS via `@theme` and `light/dark` variable blocks. Copy `src/index.css` verbatim; the values below are exactly what ships. The radius base is **0.875rem (14px)**, which yields a four-step ladder used consistently across the app.

2.1 Color tokens (light / dark)

Swatch	Token / Value	Usage
	<code>--background (0 0% 100%)</code>	Page canvas in panels; app canvas is zinc-100 (see 3.1).

	--foreground (224 71% 4%)	Body text (near-black).
	--primary (220 13% 14%)	Buttons / primary actions — near-black, not colored.
	--secondary (220 14% 96%)	Secondary surfaces, hover fills.
	--border (220 13% 91%)	1px borders everywhere.
	--muted-foreground (220 9% 46%)	Hint / secondary text.
	--destructive (0 72% 51%)	Errors / destructive actions.
	--sidebar (220 14% 96%)	Sidebar background (zinc-100).

2.2 Radius ladder

Token	Value & where it is used
--radius-sm	10px (--radius - 4px) → sidebar items, chips
--radius-md	12px (--radius - 2px) → buttons
--radius-lg	14px (--radius) → default radius
--radius-xl	18px (--radius + 4px) → cards, panels
--radius-2xl	22px (dialogs / modals)

2.3 The full token block (paste into src/index.css)

```
src/index.css (tokens only)
@import "tailwindcss";
@import "tw-animate-css";

@custom-variant dark (&:is(.dark *));

:root {
  --background: 0 0% 100%;
  --foreground: 224 71% 4%;
  --card: 0 0% 100%;
  --card-foreground: 224 71% 4%;
  --popover: 0 0% 100%;
  --popover-foreground: 224 71% 4%;
  --primary: 220 13% 14%;
  --primary-foreground: 210 20% 98%;
  --secondary: 220 14% 96%;
  --secondary-foreground: 220 13% 14%;
  --muted: 220 14% 96%;
  --muted-foreground: 220 9% 46%;
  --accent: 220 14% 96%;
  --accent-foreground: 220 13% 14%;
  --destructive: 0 72% 51%;
  --destructive-foreground: 210 20% 98%;
  --border: 220 13% 91%;
  --input: 220 13% 91%;
  --ring: 220 13% 14%;
  --sidebar: 220 14% 96%;
  --sidebar-foreground: 220 13% 14%;
  --sidebar-border: 220 13% 91%;
  --radius: 0.875rem;
  color-scheme: light;
}
```

```
.dark {
  --background: 224 71% 4%;
  --foreground: 210 20% 98%;
  --card: 224 71% 6%;
  --popover: 224 71% 6%;
  --primary: 210 20% 98%;
  --primary-foreground: 220 13% 14%;
  --secondary: 217 33% 17%;
  --muted: 217 33% 17%;
  --muted-foreground: 215 20% 65%;
  --accent: 217 33% 17%;
  --border: 217 33% 17%;
  --input: 217 33% 17%;
  --ring: 213 27% 84%;
  --sidebar: 217 33% 17%;
  --sidebar-foreground: 210 20% 98%;
  --sidebar-border: 217 33% 22%;
  color-scheme: dark;
}

@theme inline {
  --color-background: hsl(var(--background));
  --color-foreground: hsl(var(--foreground));
  --color-card: hsl(var(--card));
  --color-primary: hsl(var(--primary));
  --color-secondary: hsl(var(--secondary));
  --color-muted: hsl(var(--muted));
  --color-muted-foreground: hsl(var(--muted-foreground));
  --color-accent: hsl(var(--accent));
  --color-destructive: hsl(var(--destructive));
  --color-border: hsl(var(--border));
  --color-input: hsl(var(--input));
  --color-ring: hsl(var(--ring));
  --color-sidebar: hsl(var(--sidebar));
  --color-sidebar-foreground: hsl(var(--sidebar-foreground));
  --color-sidebar-border: hsl(var(--sidebar-border));
  --radius-sm: calc(var(--radius) - 4px);
  --radius-md: calc(var(--radius) - 2px);
  --radius-lg: var(--radius);
  --radius-xl: calc(var(--radius) + 4px);
}
```

3. Surfaces & color conventions

3.1 The canvas is gray, panels are white

The signature look is **floating panels on a light-gray canvas**. The app shell sets `bg-zinc-100 p-2` and stacks a 240px sidebar next to a white content area. Individual content panels use `bg-white / bg-zinc-50` with a `1px border-zinc-200` and `shadow-sm`. Do not make the whole page white — that flattens the depth that defines the style.

AppLayout shell (verbatim pattern)

```
<div className="flex h-screen gap-2 overflow-hidden bg-zinc-100 p-2 text-zinc-950">
  <aside className="w-[var(--sidebar-width)] ..." ... </aside>
  <main className="relative min-w-0 flex-1 overflow-hidden">
    <div className="h-full overflow-y-auto rounded-xl border border-zinc-200/80 bg-zinc-50">
      { /* white cards float here */ }
    </div>
  </main>
</div>
```

3.2 Semantic accents are used sparingly

Reserve color for meaning, not decoration. Success uses `emerald-50/200`, errors and low-balance warnings use `red-50/200`, everything else stays zinc. The low-credits badge is the canonical example of conditional accent coloring.

Conditional accent (credits badge)

```
<Badge variant="outline" className={cn(
  'px-1.5 py-0 text-[10px] font-medium tabular-nums',
  isLow ? 'bg-red-50 text-red-600 border-red-200'
    : 'bg-zinc-50 text-zinc-600 border-zinc-200')}>
  {balance}
</Badge>
```

4. The class helper & variant pattern

Every component composes classes through `cn( )` (`clsx + tailwind-merge`) and expresses variants with `cva`. This is what makes `shadcn` components themeable and conflict-free. Re-create `src/lib/utils.ts` exactly.

src/lib/utils.ts

```
// src/lib/utils.ts
import { type ClassValue, clsx } from 'clsx'
import { twMerge } from 'tailwind-merge'

export function cn(...inputs: ClassValue[]) {
  return twMerge(clsx(inputs))
}
```

Buttons follow the `shadcn new-york` contract: a default (near-black) variant, plus `outline / secondary / ghost / link`, and a size scale (`xs, sm, default, lg, icon*`). Note buttons use `rounded-md` (12px) while cards use `rounded-xl` (18px).

5. Component library

5.1 Initialize shadcn/ui with these exact settings

The `components.json` drives every generator run. Match it so future `npx shadcn add` commands stay consistent.

components.json

```
{
  "style": "new-york",
```

```
"rsc": false,
"tsx": true,
"tailwind": { "config": "", "css": "src/index.css",
               "baseColor": "neutral", "cssVariables": true, "prefix": "" },
"aliases": { "components": "@components", "ui": "@components/ui",
             "hooks": "@hooks", "lib": "@lib", "utils": "@lib/utils" },
"iconLibrary": "lucide"
}
```

5.2 Installed component inventory

These are the primitives in `src/components/ui`. Re-add them with the CLI rather than hand-writing, to inherit the new-york neutral styling.

Group	Components
Layout / display	card, badge, avatar, separator, skeleton, spinner, sheet, collapsible, sidebar
Forms	button, input, textarea, label, field, select, switch, radio-group, tabs, progress
Overlay	tooltip, (dialogs via radix-ui where needed)
AI-specific	ai-elements/chain-of-thought (custom markdown/thought stream)

6. Typography

Body uses Inter with antialiasing. The base text color is zinc-950 on the zinc-100 canvas and zinc-900 inside white panels. Headings are font-**semibold** with tight leading; descriptions and hints use text-muted-foreground.

Global typography (index.css)

```
body {
  @apply bg-background text-foreground antialiased;
  font-family: Inter, ui-sans-serif, system-ui, -apple-system,
    BlinkMacSystemFont, "Segoe UI", sans-serif;
  margin: 0;
}
```

Rule	Value
Typeface	Inter (weights 400/500/600; load via @fontsource/inter or Google Fonts).
Base size	14px body (text-sm on controls), 16px comfortable reading blocks.
Headings	font- semibold , leading-none / tight.
Buttons	text-sm font-medium.
Numbers	tabular-nums for balances, counters, metrics.
Muted text	text-zinc-400 / 500 / 600 (= --muted-foreground).

7. Layout shell

7.1 Sidebar

The left nav is a fixed 240px column on zinc-100. Items are 36px tall, rounded-sm, with an 18px lucide icon at strokeWidth 1.75. Active state is bg-zinc-200/80 text-black; inactive is text-black hover:bg-zinc-200/70. Note labels are pure black (text-black), not the muted foreground — this keeps the nav legible and crisp.

SidebarItem (verbatim)

```
const classes = (isActive: boolean) => cn(
  'flex h-9 w-full items-center gap-2 rounded-sm px-3 text-sm transition-colors',
  isActive ? 'bg-zinc-200/80 text-black' : 'text-black hover:bg-zinc-200/70')
// icon:
<Icon className="size-[18px] shrink-0" strokeWidth={1.75} />
```

7.2 Sticky header with scroll-aware divider

The 56px (h-14) sticky header sits on zinc-100. A 1px hairline with a soft shadow fades in only after the nav scrolls away from the top (opacity toggled by a scroll listener). This subtle detail is part of the polish — replicate it rather than always showing a border.

8. Iconography

Use **lucide-react** exclusively. The house style is thin, precise line icons: size-[18px] in navigation, size-4 inline with buttons, and a consistent strokeWidth 1.75 (slightly thinner than the default 2) for nav icons. Buttons automatically size their icons via [&_svg:not([class*= 'size- '])]:size-4.

Context	Spec
Nav icons	size-[18px], strokeWidth 1.75
Inline / button icons	size-4 (16px), default stroke
Tiny controls	size-3.5 (e.g. the sidebar switcher)

Brand mark

Keep monochrome; avoid colorful glyphs.

9. Cards & panels

Cards use `rounded-xl border bg-card shadow-sm` with 24px horizontal padding (`px-6`) and 24px vertical rhythm (`py-6, gap-6`). The shadow Card ships header / title / description / action / content / footer sub-components — use them rather than ad-hoc divs so spacing stays uniform.

Card usage

```
<Card>
  <CardHeader>
    <CardTitle>Plan</CardTitle>
    <CardDescription>Choose a tier</CardDescription>
  </CardHeader>
  <CardContent> ... </CardContent>
</Card>
```

Entrance motion for stacked cards uses a pure-CSS keyframe (`pay-fade-up`) with a soft cubic-bezier ease — reuse it for any card grid so panels rise into place instead of popping.

10. State, data & internationalization

10.1 Server state

Wrap the app in `QueryClientProvider` and centralize keys in `queryKeys.ts`. Components call `useXxxQuery` hooks; loading states render `<Skeleton/>`, error states render a muted `text-zinc-400` message. This loading/error discipline is part of the visual consistency.

10.2 UI state

Ephemeral UI state (dialogs, toggles, local panels) lives in **Zustand** stores under `src/stores`. Keep it out of React Query.

10.3 i18n

Default language is **English** (global product). Chinese is the secondary locale. Language is persisted in `localStorage[ 'lemma-lang' ]`, detected from storage then navigator, and synced to `<html lang>`. Keys are flat and feature-prefixed (`nav.*`, `auth.*`, `learnSpace.*`). Copy the i18n bootstrap so language labels render in their own tongue (English / Chinese (Simplified)).

11. Motion & micro-polish

Motion is restrained. Animations come from **tw-animate-css** (the Tailwind v4 successor to `tailwindcss-animate`) plus a few hand-written keyframes. Two custom scrollbar utilities are part of the signature feel and should be carried over.

11.1 Custom utilities (paste into `index.css`)

Scrollbar utilities

```
@utility scrollbar-hidden {
  scrollbar-width: none;
  -ms-overflow-style: none;
  &::-webkit-scrollbar { display: none; }
}

@utility scrollbar-fade {
  scrollbar-width: thin;
  scrollbar-color: transparent transparent;
  &:hover, &:focus-within {
    scrollbar-color: hsl(var(--muted-foreground) / 0.45) transparent;
  }
  /* thumb hidden until hover, then a thin zinc track */
}
```

11.2 Entrance keyframe

pay-fade-up

```
@keyframes pay-fade-up {
  from { opacity: 0; transform: translateY(10px); }
  to   { opacity: 1; transform: translateY(0); }
}

.pay-fade-up {
  animation: pay-fade-up 0.5s cubic-bezier(0.16, 1, 0.3, 1) both;
}
```

12. Special surfaces (copy the integration, not just styling)

Surface	Implementation note
Infinite canvas	tldraw v5 embedded as a feature surface (whiteboard / board).
Course video	@vidstack/react player themed via [data-slot="course-video-player"] CSS variables — black bg, white controls, pill buttons.
Markdown / AI output	streamdown with CJK, code, math (KaTeX) and mermaid extensions; override sticky code-block actions to scroll with content.
Rich text notes	Tiptap 3 (StarterKit + Link + Placeholder), 14-item toolbar, Ctrl/Cmd+S dirty-check, fail-open editor.
Toasts	sonner mounted once at app root.

13. Build & tooling parity

Tool	Setting
Bundler	Vite 8, @vitejs/plugin-react 6.
CSS	@tailwindcss/vite (no PostCSS config needed).
Lint	ESLint 9 flat config; react-hooks + react-refresh plugins.
Tests	Vitest 3 (component + hook tests).

Type check	tsc -b && vite build in the build script.
Aliases	@/* -> src/* in both tsconfig and vite.

14. File-by-file replication checklist

Carry these files / settings over to the new project in order:

Checklist

1. package.json -> install the dependency set (Section 1)
2. vite.config.ts -> react() + tailwindcss(); '@' alias
3. tsconfig*.json -> path alias '@/*'
4. components.json -> new-york / neutral / cssVariables (Section 5.1)
5. src/index.css -> tokens + utilities + base typography (Section 2.3, 11)
6. src/lib/utils.ts -> cn() helper (Section 4)
7. src/components/ui/* -> npx shadcn add <list> (Section 5.2)
8. src/layouts/AppLayout.tsx -> shell + sidebar (Sections 3.1, 7)
9. src/components/SidebarItem.tsx -> nav item spec (Section 7.1)
10. src/i18n/* -> i18next bootstrap, en default (Section 10.3)
11. src/lib/queryClient.ts + queryKeys.ts -> TanStack setup (Section 10.1)
12. src/stores/* -> Zustand UI stores (Section 10.2)

Golden rule: keep the canvas gray, the panels white, the primary near-black, the radii generous (14–18px), the icons thin (1.75), and the motion subtle. Match those six properties and any React app will read as the same design language.